

```

>
>
>
>
>
>
> a:=matrix(4,4,[.7,-1.7,-2.9,.5,-1.7,.7,-.5,2.9,-2.4,2.4,2.2,-2.2,
-2.4,2.4,-.2,.2]);z:=matrix(4,1,[1,-1,0,0]);puiss(a,z,10,34);

```

$$a := \begin{bmatrix} 0.7 & -1.7 & -2.9 & 0.5 \\ -1.7 & 0.7 & -.5 & 2.9 \\ -2.4 & 2.4 & 2.2 & -2.2 \\ -2.4 & 2.4 & -.2 & 0.2 \end{bmatrix} \quad (1)$$

$$z := \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$k = , 0, z = , \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.7071067814 \\ -.7071067814 \\ 0. \\ 0. \end{bmatrix}, lambda = , 0$$

$$k = , 1, z = , \begin{bmatrix} 1.697056275 \\ -1.697056275 \\ -3.394112550 \\ -3.394112550 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3162277660028180775 \\ -.3162277660028180775 \\ -.6324555320056361550 \\ -.6324555320056361550 \end{bmatrix}, lambda = , 2.400000000$$

$$k = , 2, z = , \begin{bmatrix} 2.276839915 \\ -2.276839915 \\ -1.517893277 \\ -1.517893277 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5883484054206797465 \\ -.5883484054206797465 \\ -.3922322703665883967 \\ -.3922322703665883967 \end{bmatrix}, lambda = , 3.360000000$$

$$k = , 3, z = , \begin{bmatrix} 2.353393622 \\ -2.353393622 \\ -2.824072346 \\ -2.824072346 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4526787302111839304 \\ -.4526787302111839304 \\ -.5432144761764801272 \\ -.5432144761764801272 \end{bmatrix}, lambda = , 4.984615386$$

$k = , 4, z = ,$	$\begin{bmatrix} 2.390143695 \\ -2.390143695 \\ -2.172857904 \\ -2.172857904 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.5232166436845309980 \\ -.5232166436845309980 \\ -.4756514941392613056 \\ -.4756514941392613056 \end{bmatrix}$	$, lambda = , 4.524590162$
$k = , 5, z = ,$	$\begin{bmatrix} 2.397283531 \\ -2.397283531 \\ -2.511439890 \\ -2.511439890 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.4882400826440017032 \\ -.4882400826440017032 \\ -.5114896104665404080 \\ -.5114896104665404080 \end{bmatrix}$	$, lambda = , 4.897737558$
$k = , 6, z = ,$	$\begin{bmatrix} 2.399351263 \\ -2.399351263 \\ -2.343552396 \\ -2.343552396 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.5058473475486774343 \\ -.5058473475486774343 \\ -.4940834556569667756 \\ -.4940834556569667756 \end{bmatrix}$	$, lambda = , 4.740324322$
$k = , 7, z = ,$	$\begin{bmatrix} 2.399833928 \\ -2.399833928 \\ -2.428067268 \\ -2.428067268 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.4970675238840261496 \\ -.4970675238840261496 \\ -.5029153770379614876 \\ -.5029153770379614876 \end{bmatrix}$	$, lambda = , 4.827234986$
$k = , 8, z = ,$	$\begin{bmatrix} 2.399958962 \\ -2.399958962 \\ -2.385924114 \\ -2.385924114 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.5014641193948237758 \\ -.5014641193948237758 \\ -.4985315806287045726 \\ -.4985315806287045726 \end{bmatrix}$	$, lambda = , 4.785719168$
$k = , 9, z = ,$	$\begin{bmatrix} 2.399989681 \\ -2.399989681 \\ -2.407027774 \\ -2.407027774 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.4992674005489760180 \\ -.4992674005489760180 \\ -.5007315278428349720 \\ -.5007315278428349720 \end{bmatrix}$	$, lambda = , 4.806976146$
$k = , 10, z = ,$	$\begin{bmatrix} 2.399997428 \\ -2.399997428 \\ -2.396483522 \\ -2.396483522 \end{bmatrix}$	$, z\ norme = ,$	$\begin{bmatrix} 0.5003661661736839280 \\ -.5003661661736839280 \\ -.4996335655246157720 \\ -.4996335655246157720 \end{bmatrix}$	$, lambda = ,$
4.796470664				

$$k = , 11, z = , \begin{bmatrix} 2.399999356 \\ -2.399999356 \\ -2.401757598 \\ -2.401757598 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4998168833224336536 \\ -.4998168833224336536 \\ -.5001830496859243788 \\ -.5001830496859243788 \end{bmatrix}, \lambda = ,$$

4.801754378

$$k = , 12, z = , \begin{bmatrix} 2.399999839 \\ -2.399999839 \\ -2.399121040 \\ -2.399121040 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000915498921896109 \\ -.5000915498921896109 \\ -.4999084332324248240 \\ -.4999084332324248240 \end{bmatrix}, \lambda = ,$$

4.799120234

$$k = , 13, z = , \begin{bmatrix} 2.399999959 \\ -2.399999959 \\ -2.400439440 \\ -2.400439440 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999542226591152170 \\ -.4999542226591152170 \\ -.5000457728197327200 \\ -.5000457728197327200 \end{bmatrix}, \lambda = ,$$

4.800439236

$$k = , 14, z = , \begin{bmatrix} 2.399999989 \\ -2.399999989 \\ -2.399780268 \\ -2.399780268 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000228881082284190 \\ -.5000228881082284190 \\ -.4999771108042694280 \\ -.4999771108042694280 \end{bmatrix}, \lambda = ,$$

4.799780216

$$k = , 15, z = , \begin{bmatrix} 2.399999998 \\ -2.399999998 \\ -2.400109862 \\ -2.400109862 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999885560633428696 \\ -.4999885560633428696 \\ -.5000114438728300024 \\ -.5000114438728300024 \end{bmatrix}, \lambda = ,$$

4.800109848

$$k = , 16, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.399945070 \\ -2.399945070 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000057220000000000 \\ -.5000057220000000000 \\ -.4999942781190377250 \\ -.4999942781190377250 \end{bmatrix}, \lambda = ,$$

4.799945070

$$k = , 17, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.400027466 \\ -2.400027466 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999971388800000000 \\ -.4999971388800000000 \\ -.5000028609305901992 \\ -.5000028609305901992 \end{bmatrix}, \lambda = ,$$

4.800027466

$$k = , 18, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.399986266 \\ -2.399986266 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000014305600000000 \\ -.5000014305600000000 \\ -.4999985693018136204 \\ -.4999985693018136204 \end{bmatrix}, \lambda = ,$$

4.799986264

$$k = , 19, z = , \begin{bmatrix} 2.399999999 \\ -2.399999999 \\ -2.400006866 \\ -2.400006866 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999992845116669647 \\ -.4999992845116669647 \\ -.5000007151346203698 \\ -.5000007151346203698 \end{bmatrix}, \lambda = ,$$

4.800006864

$$k = , 20, z = , \begin{bmatrix} 2.399999999 \\ -2.399999999 \\ -2.399996566 \\ -2.399996566 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000003575516665176 \\ -.5000003575516665176 \\ -.4999996423428214384 \\ -.4999996423428214384 \end{bmatrix}, \lambda = ,$$

4.799996562

$$k = , 21, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.400001716 \\ -2.400001716 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999998211200000000 \\ -.4999998211200000000 \\ -.5000001786198721008 \\ -.5000001786198721008 \end{bmatrix}, \lambda = ,$$

4.800001716

$$k = , 22, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.399999142 \\ -2.399999142 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000894400000000 \\ -.5000000894400000000 \\ -.4999999106899680252 \\ -.4999999106899680252 \end{bmatrix}, \lambda = ,$$

4.799999142

$$k = , 23, z = , \begin{bmatrix} 2.400000001 \\ -2.400000001 \\ -2.400000430 \\ -2.400000430 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999999554883333147 \\ -.4999999554883333147 \\ -.5000000448633253210 \\ -.5000000448633253210 \end{bmatrix}, \lambda = ,$$

4.800000432

$$k = , 24, z = , \begin{bmatrix} 2.400000001 \\ -2.400000001 \\ -2.399999786 \\ -2.399999786 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000224483333426 \\ -.5000000224483333426 \\ -.4999999776566646836 \\ -.4999999776566646836 \end{bmatrix}, \lambda = ,$$

4.799999790

$$k = , 25, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.400000108 \\ -2.400000108 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999999886400000000 \\ -.4999999886400000000 \\ -.5000000111399994888 \\ -.5000000111399994888 \end{bmatrix}, \lambda = ,$$

4.800000108

$$k = , 26, z = , \begin{bmatrix} 2.399999999 \\ -2.399999999 \\ -2.399999946 \\ -2.399999946 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000054716666643 \\ -.5000000054716666643 \\ -.4999999944299998722 \\ -.4999999944299998722 \end{bmatrix}, \lambda = ,$$

4.799999944

$$k = , 27, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.400000026 \\ -2.400000026 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999999972800000000 \\ -.4999999972800000000 \\ -.5000000026966666372 \\ -.5000000026966666372 \end{bmatrix}, \lambda = ,$$

4.800000026

$$k = , 28, z = , \begin{bmatrix} 2.400000001 \\ -2.400000001 \\ -2.399999988 \\ -2.399999988 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000015683333339 \\ -.5000000015683333339 \\ -.4999999988599999932 \\ -.4999999988599999932 \end{bmatrix}, \lambda = ,$$

4.799999988

$$k = , 29, z = , \begin{bmatrix} 2.400000002 \\ -2.400000002 \\ -2.400000008 \\ -2.400000008 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999999993766666658 \\ -.4999999993766666658 \\ -.5000000006266666632 \\ -.5000000006266666632 \end{bmatrix}, \lambda = ,$$

4.800000012

$$k = , 30, z = , \begin{bmatrix} 2.400000001 \\ -2.400000001 \\ -2.399999998 \\ -2.399999998 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000003683333334 \\ -.5000000003683333334 \\ -.4999999997433333332 \\ -.4999999997433333332 \end{bmatrix}, \lambda = ,$$

4.799999998

$$k = , 31, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.400000002 \\ -2.400000002 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999999996800000000 \\ -.4999999996800000000 \\ -.5000000000966666664 \\ -.5000000000966666664 \end{bmatrix}, \lambda = ,$$

4.800000002

$$k = , 32, z = , \begin{bmatrix} 2.399999999 \\ -2.399999999 \\ -2.399999998 \\ -2.399999998 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000001916666665 \\ -.5000000001916666665 \\ -.4999999998333333330 \\ -.4999999998333333330 \end{bmatrix}, \lambda = ,$$

4.799999996

$$k = , 33, z = , \begin{bmatrix} 2.400000000 \\ -2.400000000 \\ -2.400000000 \\ -2.400000000 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4999999992000000000 \\ -.4999999992000000000 \\ -.4999999992000000000 \\ -.4999999992000000000 \end{bmatrix}, \lambda = ,$$

4.800000000

4.800000000

```
> a:=matrix(4,4,[.7,-1.7,-2.9,.5,-1.7,.7,-.5,2.9,-2.4,2.4,2.2,-2.2,
-2.4,2.4,-.2,.2]);z:=matrix(4,1,[0,0,1,1]);puiss_inverses(a,z,10,
50);
```

(2)

$$a := \begin{bmatrix} 0.7 & -1.7 & -2.9 & 0.5 \\ -1.7 & 0.7 & -.5 & 2.9 \\ -2.4 & 2.4 & 2.2 & -2.2 \\ -2.4 & 2.4 & -.2 & 0.2 \end{bmatrix}$$

$$z := \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$k = , 0, z = , \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0. \\ 0. \\ 0.7071067814 \\ 0.7071067814 \end{bmatrix}, lambda = , 0$$

$$k = , 1, z = , \begin{bmatrix} -.1473139131 \\ 0.1473139121 \\ -.1473139125 \\ -.1473139128 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.50000000016842611443 \\ 0.4999999982901485913 \\ -.4999999996477936125 \\ -.50000000006660273784 \end{bmatrix}, lambda = , \\ -4.8000000003$$

$$k = , 2, z = , \begin{bmatrix} 0.1041666687 \\ -.1041666649 \\ 0.3125000000 \\ 0.3124999998 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2236068020731380246 \\ -.2236067939159620442 \\ 0.6708203931250000000 \\ 0.6708203926956749484 \end{bmatrix}, lambda = , \\ -2.3999999999$$

$$k = , 3, z = , \begin{bmatrix} -.1397542504 \\ 0.1397542493 \\ -.2329237500 \\ -.2329237477 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3638034400450476912 \\ 0.3638034371815656654 \\ -.6063390650063025000 \\ -.6063390590190219006 \end{bmatrix}, lambda = , \\ -2.6666666652$$

$$k = , 4, z = , \begin{bmatrix} 0.1263206457 \\ -.1263206268 \\ 0.2779054000 \\ 0.2779054047 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2926028876695017075 \\ -.2926028438904777300 \\ 0.6437263052950650000 \\ 0.6437263161819122325 \end{bmatrix}, lambda = ,$$

−2.331428587

$$k = , 5, z = , \begin{bmatrix} -.1341096594 \\ 0.1341096414 \\ -.2560275150 \\ -.2560275075 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3281025176540716626 \\ 0.3281024736166373406 \\ -.6263774931354094350 \\ -.6263774747864784675 \end{bmatrix}, \lambda = ,$$

−2.450349628

$$k = , 6, z = , \begin{bmatrix} 0.1304953453 \\ -.1304952682 \\ 0.2672046750 \\ 0.2672046845 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3103035032970329414 \\ -.3103033199617572716 \\ 0.6353831744667226500 \\ 0.6353831970566723110 \end{bmatrix}, \lambda = ,$$

−2.378836001

$$k = , 7, z = , \begin{bmatrix} -.1323715771 \\ 0.1323714191 \\ -.2616645900 \\ -.2616645825 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3191937165040254659 \\ 0.3191933355113054839 \\ -.6309639485257901100 \\ -.6309639304406926425 \end{bmatrix}, \lambda = ,$$

−2.411584024

$$k = , 8, z = , \begin{bmatrix} 0.1314510200 \\ -.1314506204 \\ 0.2644481200 \\ 0.2644481255 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3147462188360472600 \\ -.3147452620341217452 \\ 0.6331943704073295600 \\ 0.6331943835765252315 \end{bmatrix}, \lambda = ,$$

−2.394458493

$$k = , 9, z = , \begin{bmatrix} -.1319159687 \\ 0.1319150215 \\ -.2630595550 \\ -.2630595521 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3169701355782516354 \\ 0.3169678596285286530 \\ -.6320843764042688100 \\ -.6320843694360950382 \end{bmatrix}, \lambda = ,$$

−2.402833380

$$k = , 10, z = , \begin{bmatrix} 0.1316853843 \\ -.1316831053 \\ 0.2637546600 \\ 0.2637546609 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3158599379348314965 \\ -.3158544715362453515 \\ 0.6326406759601383000 \\ 0.6326406781188740295 \end{bmatrix}, \lambda = ,$$

−2.398598962

$$k = , 11, z = , \begin{bmatrix} -.1318028729 \\ 0.1317974090 \\ -.2634073100 \\ -.2634073095 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3164196186764937687 \\ 0.3164065014726237270 \\ -.6323628518328069300 \\ -.6323628506324551785 \end{bmatrix}, \lambda = ,$$

−2.400704429

$$k = , 12, z = , \begin{bmatrix} 0.1317488200 \\ -.1317357005 \\ 0.2635810350 \\ 0.2635810359 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3161508629009787000 \\ -.3161193807120237675 \\ 0.6325018444915337250 \\ 0.6325018466512174065 \end{bmatrix}, \lambda = ,$$

−2.399648766

$$k = , 13, z = , \begin{bmatrix} -.1317869593 \\ 0.1317554763 \\ -.2634941850 \\ -.2634941849 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3163118709945927772 \\ 0.3162363062597554452 \\ -.6324323673316997400 \\ -.6324323670916821596 \end{bmatrix}, \lambda = ,$$

−2.400175864

$$k = , 14, z = , \begin{bmatrix} 0.1317945236 \\ -.1317189643 \\ 0.2635376150 \\ 0.2635376134 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3162952705834303240 \\ -.3161139349058483870 \\ 0.6324671084082653500 \\ 0.6324671045684060060 \end{bmatrix}, \lambda = ,$$

−2.399912112

$$k = , 15, z = , \begin{bmatrix} -.1318546473 \\ 0.1316733158 \\ -.2635159000 \\ -.2635158988 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3164569341595922793 \\ 0.3160217306098379878 \\ -.6324497128005719000 \\ -.6324497099205192908 \end{bmatrix}, \lambda = ,$$

−2.400043884

$$k = , 16, z = , \begin{bmatrix} 0.1319779591 \\ -.1315427542 \\ 0.2635267450 \\ 0.2635267455 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3167441300922949453 \\ -.3156996481318036786 \\ 0.6324582541682828350 \\ 0.6324582553682715765 \end{bmatrix}, \lambda = ,$$

−2.399977707

$$\begin{aligned}
& k = , 17, z = , \begin{bmatrix} -.1322843737 \\ 0.1312399023 \\ -.2635212600 \\ -.2635212568 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3174835358414710398 \\ 0.3149767962781926642 \\ -.6324530936959760400 \\ -.6324530860159509072 \end{bmatrix}, \lambda = , \\
& \quad -2.400009139 \\
& k = , 18, z = , \begin{bmatrix} 0.1330144383 \\ -.1305076792 \\ 0.2635236250 \\ 0.2635236306 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3192315361897972608 \\ -.3132153730681224192 \\ 0.6324505272226080000 \\ 0.6324505406624768256 \end{bmatrix}, \lambda = , \\
& \quad -2.399983967 \\
& k = , 19, z = , \begin{bmatrix} -.1347686010 \\ 0.1287524577 \\ -.2635203050 \\ -.2635202990 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3234310810399871730 \\ 0.3089929425064390221 \\ -.6324222147422687650 \\ -.6324222003428725270 \end{bmatrix}, \lambda = , \\
& \quad -2.399941933 \\
& k = , 20, z = , \begin{bmatrix} 0.1389737084 \\ -.1245355405 \\ 0.2635096250 \\ 0.2635096335 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.3334540027598605252 \\ -.2988110121236983715 \\ 0.6322659172992163750 \\ 0.6322659376941461505 \end{bmatrix}, \lambda = , \\
& \quad -2.399648485 \\
& k = , 21, z = , \begin{bmatrix} -.1490435471 \\ 0.1144005952 \\ -.2634439550 \\ -.2634439438 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3571948440244239342 \\ 0.2741702244334552704 \\ -.6313646195783699100 \\ -.6313645927366694076 \end{bmatrix}, \lambda = , \\
& \quad -2.397986127 \\
& k = , 22, z = , \begin{bmatrix} 0.1730466186 \\ -0.09002196340 \\ 0.2630686750 \\ 0.2630686854 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4119464677717889304 \\ -.21430196177500611760 \\ 0.6262486509381277000 \\ 0.6262486756958681256 \end{bmatrix}, \lambda = , \\
& \quad -2.388474887
\end{aligned}$$

$$\begin{aligned}
& k = , 23, z = , \begin{bmatrix} -.2292907049 \\ 0.03164624569 \\ -.2609369050 \\ -.2609368913 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.5263728121972007098 \\ 0.07264892550526082138 \\ -.5990216330434588100 \\ -.5990216015929578826 \end{bmatrix}, \lambda = , \\
& \quad -2.336120473 \\
& k = , 24, z = , \begin{bmatrix} 0.3516581291 \\ 0.1020657902 \\ 0.2495923600 \\ 0.2495923694 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.6914243006566373126 \\ 0.2006800405570435772 \\ 0.4907443015860589600 \\ 0.4907443200681809084 \end{bmatrix}, \lambda = , \\
& \quad -2.097708086 \\
& k = , 25, z = , \begin{bmatrix} -.5482905561 \\ -.3438137558 \\ -.2044767920 \\ -.2044767833 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.7735056661147169529 \\ -.4850382433925741462 \\ -.2884674110128456880 \\ -.2884673987392450937 \end{bmatrix}, \lambda = , \\
& \quad -1.541331704 \\
& k = , 26, z = , \begin{bmatrix} 0.6893693384 \\ 0.5691745887 \\ 0.1201947520 \\ 0.1201947573 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.7575563369796361912 \\ 0.6254728670094605841 \\ 0.1320834724976735360 \\ 0.1320834783219078939 \end{bmatrix}, \lambda = , \\
& \quad -1.138113395 \\
& k = , 27, z = , \begin{bmatrix} -.7190319900 \\ -.6639972068 \\ -0.05503478200 \\ -0.05503477958 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.7323511940573300100 \\ -.6762969575953246332 \\ -0.05605423523977681800 \\ -0.05605423277494922442 \end{bmatrix}, \lambda = , \\
& \quad -1.026106355 \\
& k = , 28, z = , \begin{bmatrix} 0.7160020413 \\ 0.6926461126 \\ 0.02335593050 \\ 0.02335593155 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.7183373592859204104 \\ 0.6949052527020250608 \\ 0.02343210838976624400 \\ 0.02343210944319093240 \end{bmatrix}, \lambda = , \\
& \quad -1.004603287
\end{aligned}$$

$$k = , 29, z = , \begin{bmatrix} -.7115029939 \\ -.7017396160 \\ -0.009763378400 \\ -0.009763377981 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.7119068856895172740 \\ -.7021379655104185600 \\ -0.009768920679382544000 \\ -0.009768920260144694460 \end{bmatrix}, lambda = ,$$

—1.000801362

$$k = , 30, z = , \begin{bmatrix} 0.7090576176 \\ 0.7049872339 \\ 0.004070383350 \\ 0.004070383481 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.7091275264267496544 \\ 0.7050567414113391366 \\ 0.004070784665376009900 \\ 0.004070784796388925714 \end{bmatrix}, lambda = ,$$

—1.000139190

$$k = , 31, z = , \begin{bmatrix} -.7079402151 \\ -.7062440533 \\ -0.001696160630 \\ -0.001696160552 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.7079523336206020818 \\ -.7062561427857043894 \\ -0.001696189664877664340 \\ -0.001696189586876329136 \end{bmatrix}, lambda = ,$$

—1.000024167

$$k = , 32, z = , \begin{bmatrix} 0.7074576106 \\ 0.7067508652 \\ 0.0007067459450 \\ 0.0007067459731 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.7074597131640187032 \\ 0.7067529656635713744 \\ 0.0007067480454489485400 \\ 0.0007067480735490320532 \end{bmatrix}, lambda = ,$$

—1.000004196

$$k = , 33, z = , \begin{bmatrix} -.7072535797 \\ -.7069591007 \\ -0.0002944781450 \\ -0.0002944781339 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.7072539439355935455 \\ -.7069594647839368605 \\ -0.0002944782966562446750 \\ -0.0002944782855562389585 \end{bmatrix}, lambda = ,$$

—1.000000728

$$k = , 34, z = , \begin{bmatrix} 0.7071680530 \\ 0.7070453547 \\ 0.0001226994325 \\ 0.0001226994370 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.7071681166451247700 \\ 0.7070454183340819230 \\ 0.0001226994435429489250 \\ 0.0001226994480429493300 \end{bmatrix}, lambda = ,$$

—1.000000128

$$k = , 35, z = , \begin{bmatrix} -.7071323306 \\ -.7070812046 \\ -0.00005112467750 \\ -0.00005112467566 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.7071323419141172896 \\ -.7070812159132992736 \\ -0.00005112467831799484000 \\ -0.00005112467647799481056 \end{bmatrix},$$

$$lambda = , -1.0000000023$$

$$k = , 36, z = , \begin{bmatrix} 0.7071174301 \\ 0.7070961280 \\ 0.00002130222365 \\ 0.00002130222469 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.7071174315142348602 \\ 0.7070961294141922560 \\ 0.00002130222369260444730 \\ 0.00002130222473260444938 \end{bmatrix}, lambda = ,$$

$$-1.0000000004$$

$$k = , 37, z = , \begin{bmatrix} -.7071112193 \\ -.7071023422 \\ -0.000008876088650 \\ -0.000008876088252 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.70711121930000000000 \\ -.70710234220000000000 \\ -0.000008876088650000000000 \\ -0.000008876088252000000000 \end{bmatrix},$$

$$lambda = , -1.0000000001$$

$$k = , 38, z = , \begin{bmatrix} 0.7071086296 \\ 0.7071049314 \\ 0.000003698768350 \\ 0.000003698768474 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.7071086303071086296 \\ 0.7071049321071049314 \\ 0.000003698768353698768350 \\ 0.000003698768477698768474 \end{bmatrix}, lambda = ,$$

$$-1.0000000002$$

$$k = , 39, z = , \begin{bmatrix} -.7071075524 \\ -.7071060108 \\ -0.000001540993455 \\ -0.000001540993405 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.70710755240000000000 \\ -.70710601080000000000 \\ -0.000001540993455000000000 \\ -0.000001540993405000000000 \end{bmatrix},$$

$$lambda = , -.9999999990$$

$$k = , 40, z = , \begin{bmatrix} 0.7071071037 \\ 0.7071064609 \\ 6.420819450\ 10^{-7} \\ 6.420819708\ 10^{-7} \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.70710710228578579260 \\ 0.70710645948578707820 \\ 6.4208194371583611000\ 10^{-7} \\ 6.4208196951583605840\ 10^{-7} \end{bmatrix}, lambda = ,$$

$$-.99999999980$$

$$k = , 41, z = , \begin{bmatrix} -.7071069153 \\ -.7071066475 \\ -2.675170800 \cdot 10^{-7} \\ -2.675170712 \cdot 10^{-7} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.70710691530000000000 \\ -.70710664750000000000 \\ -2.6751708000000000000 \cdot 10^{-7} \\ -2.6751707120000000000 \cdot 10^{-7} \end{bmatrix}, \lambda = ,$$

—1.0000000000

$$k = , 42, z = , \begin{bmatrix} 0.7071068367 \\ 0.7071067260 \\ 1.113577205 \cdot 10^{-7} \\ 1.113577252 \cdot 10^{-7} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.70710683670000000000 \\ 0.70710672600000000000 \\ 1.1135772050000000000 \cdot 10^{-7} \\ 1.1135772520000000000 \cdot 10^{-7} \end{bmatrix}, \lambda = ,$$

—1.0000000000

$$k = , 43, z = , \begin{bmatrix} -.7071068047 \\ -.7071067581 \\ -4.632452700 \cdot 10^{-8} \\ -4.632452502 \cdot 10^{-8} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.70710680470000000000 \\ -.70710675810000000000 \\ -4.6324527000000000000 \cdot 10^{-8} \\ -4.6324525020000000000 \cdot 10^{-8} \end{bmatrix}, \lambda = ,$$

—.9999999990

$$k = , 44, z = , \begin{bmatrix} 0.7071067904 \\ 0.7071067715 \\ 1.944260890 \cdot 10^{-8} \\ 1.944260983 \cdot 10^{-8} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.70710679040000000000 \\ 0.70710677150000000000 \\ 1.9442608900000000000 \cdot 10^{-8} \\ 1.9442609830000000000 \cdot 10^{-8} \end{bmatrix}, \lambda = ,$$

—1.0000000000

$$k = , 45, z = , \begin{bmatrix} -.7071067851 \\ -.7071067770 \\ -8.008877200 \cdot 10^{-9} \\ -8.008876839 \cdot 10^{-9} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.70710678510000000000 \\ -.70710677700000000000 \\ -8.0088772000000000000 \cdot 10^{-9} \\ -8.0088768390000000000 \cdot 10^{-9} \end{bmatrix}, \lambda = ,$$

—1.0000000000

$$k = , 46, z = , \begin{bmatrix} 0.7071067827 \\ 0.7071067797 \\ 3.335182600 \cdot 10^{-9} \\ 3.335182757 \cdot 10^{-9} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.70710678270000000000 \\ 0.70710677970000000000 \\ 3.3351826000000000000 \cdot 10^{-9} \\ 3.3351827570000000000 \cdot 10^{-9} \end{bmatrix}, \lambda = ,$$

—1.0000000000

$$k = , 47, z = , \begin{bmatrix} -.7071067804 \\ -.7071067808 \\ -1.111496445 \cdot 10^{-9} \\ -1.111496372 \cdot 10^{-9} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.7071067811071067804 \\ -.7071067815071067808 \\ -1.111496446111496445 \cdot 10^{-9} \\ -1.111496373111496372 \cdot 10^{-9} \end{bmatrix}, \lambda = ,$$

−1.0000000001

$$k = , 48, z = , \begin{bmatrix} 0.7071067813 \\ 0.7071067814 \\ 2.322839700 \cdot 10^{-11} \\ 2.322842752 \cdot 10^{-11} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.70710678130000000000 \\ 0.70710678140000000000 \\ 2.32283970000000000000 \cdot 10^{-11} \\ 2.32284275200000000000 \cdot 10^{-11} \end{bmatrix}, \lambda = ,$$

−1.0000000000

$$k = , 49, z = , \begin{bmatrix} -.7071067810 \\ -.7071067808 \\ -2.131725965 \cdot 10^{-10} \\ -2.131725821 \cdot 10^{-10} \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.70710678100000000000 \\ -.70710678080000000000 \\ -2.13172596500000000000 \cdot 10^{-10} \\ -2.13172582100000000000 \cdot 10^{-10} \end{bmatrix}, \lambda = ,$$

−1.0000000000

−1.0000000000

```
> a:=evalm(matrix(4,4,[.7,-1.7,-2.9,.5,-1.7,.7,-.5,2.9,-2.4,2.4,2.2,-2.2,-2.4,2.4,-.2,.2])-matrix(4,4,[2.5,0,0,0,0,2.5,0,0,0,0,2.5,0,0,0,0,2.5]));z:=matrix(4,1,[.5,.5,.5,.5]);puiss_inverses(a,z,10,50);
```

$$a := \begin{bmatrix} -1.8 & -1.7 & -2.9 & 0.5 \\ -1.7 & -1.8 & -.5 & 2.9 \\ -2.4 & 2.4 & -.3 & -2.2 \\ -2.4 & 2.4 & -.2 & -2.3 \end{bmatrix}$$

(3)

$$z := \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

$$k = , 0, z = , \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.5000000000 \\ 0.5000000000 \\ 0.5000000000 \\ 0.5000000000 \end{bmatrix}, \lambda = , 0$$

$$k = , 1, z = , \begin{bmatrix} -.2493345161 \\ -0.03637976823 \\ 0.004436556503 \\ 0.004436557999 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.9892158815841828355 \\ -.14433398578091088765 \\ 0.017601703209647131165 \\ 0.017601709144914265445 \end{bmatrix}, lambda = ,$$

-7.224359017

$$k = , 2, z = , \begin{bmatrix} 0.06447815256 \\ 0.2593932280 \\ 0.1800777976 \\ 0.1800777871 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.17464950683949963120 \\ 0.7026085200804935600 \\ 0.4877698459849177520 \\ 0.4877698175439691670 \end{bmatrix}, lambda = ,$$

-10.53932706

$$k = , 3, z = , \begin{bmatrix} -.2877533778 \\ 0.03710861281 \\ 0.1167592208 \\ 0.1167596177 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.8619676269488816952 \\ 0.11115915708010351804 \\ 0.3497532131397851072 \\ 0.3497544020570230268 \end{bmatrix}, lambda = , 11.14575129$$

$$k = , 4, z = , \begin{bmatrix} -0.07515994094 \\ 0.2896650993 \\ 0.2103412117 \\ 0.2103293249 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.17812776449589801722 \\ 0.6865012923836833659 \\ 0.4985050460775339271 \\ 0.4984768745664266987 \end{bmatrix}, lambda = ,$$

4.096421804

$$k = , 5, z = , \begin{bmatrix} -.2746033381 \\ 0.1296275503 \\ 0.1884116082 \\ 0.1886934525 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.6795063322159731255 \\ 0.3207635488626806565 \\ 0.4662247797886291110 \\ 0.4669222038908781375 \end{bmatrix}, lambda = , 3.068544350$$

$$k = , 6, z = , \begin{bmatrix} -.1609562141 \\ 0.2566792117 \\ 0.2205773984 \\ 0.2136031947 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.3731362384176813683 \\ 0.5950457773208378971 \\ 0.5113528618895012192 \\ 0.4951849360400532261 \end{bmatrix}, lambda = , 2.536276200$$

$$k = , 7, z = , \begin{bmatrix} -.2021601523 \\ 0.2958173605 \\ 0.1312394464 \\ 0.2929187897 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.4202542797260689725 \\ 0.6149506238148712875 \\ 0.2728229989490368800 \\ 0.6089250209948786775 \end{bmatrix}, lambda = , 2.156953813$$

$$\begin{aligned}
& k = , 8, z = , \begin{bmatrix} -1.153268912 \\ -2.167349565 \\ 1.875051056 \\ -1.485969040 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.3364237402154105744 \\ -.6322444309602122655 \\ 0.5469771037705517072 \\ -.4334767521081874480 \end{bmatrix}, \lambda = , \\
& \quad -.8055204261 \\
& k = , 9, z = , \begin{bmatrix} 2.833388020 \\ 6.967783179 \\ -4.877765346 \\ 4.926773191 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.27698059634985265460 \\ 0.68114240849931035067 \\ -.47683068639138432258 \\ 0.48162149585274469943 \end{bmatrix}, \lambda = , \\
& \quad -.09840361091 \\
& k = , 10, z = , \begin{bmatrix} -2.796481771 \\ -6.787945971 \\ 4.793305768 \\ -4.791216023 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2798889122332565350 \\ -.6793789374289135350 \\ 0.4797432085270382800 \\ -.4795340541313399550 \end{bmatrix}, \lambda = , \\
& \quad -.1000873065 \\
& k = , 11, z = , \begin{bmatrix} 2.797845207 \\ 6.794924705 \\ -4.796340901 \\ 4.796431701 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.27977111630754861921 \\ 0.67945991621958608615 \\ -.47961111097463353603 \\ 0.47962019053961346003 \end{bmatrix}, \lambda = , \\
& \quad -.09999521133 \\
& k = , 12, z = , \begin{bmatrix} -2.797759881 \\ -6.794553036 \\ 4.796158471 \\ -4.796154518 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797764838630509132 \\ -.6794565075947979792 \\ 0.4796166969792810612 \\ -.4796163016785805896 \end{bmatrix}, \lambda = , \\
& \quad -.1000001772 \\
& k = , 13, z = , \begin{bmatrix} 2.797762820 \\ 6.794567138 \\ -4.796164894 \\ 4.796165064 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.27977625905834487600 \\ 0.67945665808454946840 \\ -.47961645007144786920 \\ 0.47961646707144647520 \end{bmatrix}, \lambda = , \\
& \quad -.099999999180
\end{aligned}$$

$$k = , 14, z = , \begin{bmatrix} -2.797762668 \\ -6.794566479 \\ 4.796164576 \\ -4.796164568 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762684786576008 \\ -.6794566519767398874 \\ 0.4796164604776987456 \\ -.4796164596776987408 \end{bmatrix}, lambda = ,$$

-.10000000006

$$k = , 15, z = , \begin{bmatrix} 2.797762669 \\ 6.794566510 \\ -4.796164590 \\ 4.796164591 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762674595525338 \\ 0.6794566523589133020 \\ -.4796164599592329180 \\ 0.4796164600592329182 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 16, z = , \begin{bmatrix} -2.797762662 \\ -6.794566515 \\ 4.796164590 \\ -4.796164588 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762667595525324 \\ -.6794566528589133030 \\ 0.4796164599592329180 \\ -.4796164597592329176 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 17, z = , \begin{bmatrix} 2.797762648 \\ 6.794566520 \\ -4.796164583 \\ 4.796164581 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762656393287944 \\ 0.6794566540383699560 \\ -.4796164597388493749 \\ 0.4796164595388493743 \end{bmatrix}, lambda = ,$$

-.10000000003

$$k = , 18, z = , \begin{bmatrix} -2.797762637 \\ -6.794566531 \\ 4.796164583 \\ -4.796164580 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762642595525274 \\ -.6794566544589133062 \\ 0.4796164592592329166 \\ -.4796164589592329160 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 19, z = , \begin{bmatrix} 2.797762661 \\ 6.794566490 \\ -4.796164576 \\ 4.796164572 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762674988813305 \\ 0.6794566523972832450 \\ -.4796164599980822880 \\ 0.4796164595980822860 \end{bmatrix}, lambda = ,$$

-.10000000006

$$k = , 20, z = , \begin{bmatrix} -2.797762656 \\ -6.794566515 \\ 4.796164586 \\ -4.796164585 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797762664393287968 \\ -.6794566535383699545 \\ 0.4796164600388493758 \\ -.4796164599388493755 \end{bmatrix}, \lambda = ,$$

-.10000000003

$$k = , 21, z = , \begin{bmatrix} 2.797762643 \\ 6.794566526 \\ -4.796164583 \\ 4.796164582 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2797762648595525286 \\ 0.6794566539589133052 \\ -.4796164592592329166 \\ 0.4796164591592329164 \end{bmatrix}, \lambda = ,$$

-.10000000002

$$k = , 22, z = , \begin{bmatrix} -2.797762667 \\ -6.794566484 \\ 4.796164576 \\ -4.796164573 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797762683786576002 \\ -.6794566524767398904 \\ 0.4796164604776987456 \\ -.4796164601776987438 \end{bmatrix}, \lambda = ,$$

-.10000000006

$$k = , 23, z = , \begin{bmatrix} 2.797762668 \\ 6.794566510 \\ -4.796164590 \\ 4.796164588 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2797762673595525336 \\ 0.6794566523589133020 \\ -.4796164599592329180 \\ 0.4796164597592329176 \end{bmatrix}, \lambda = ,$$

-.10000000002

$$k = , 24, z = , \begin{bmatrix} -2.797762711 \\ -6.794566464 \\ 4.796164588 \\ -4.796164586 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797762722191050844 \\ -.6794566491178265856 \\ 0.4796164607184658352 \\ -.4796164605184658344 \end{bmatrix}, \lambda = ,$$

-.10000000005

$$k = , 25, z = , \begin{bmatrix} 2.797762711 \\ 6.794566474 \\ -4.796164593 \\ 4.796164595 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2797762719393288133 \\ 0.6794566494383699422 \\ -.4796164607388493779 \\ 0.4796164609388493785 \end{bmatrix}, \lambda = ,$$

-.10000000004

$$k = , 26, z = , \begin{bmatrix} -2.797762667 \\ -6.794566531 \\ 4.796164595 \\ -4.796164597 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.27977626670000000000 \\ -.67945665310000000000 \\ 0.47961645950000000000 \\ -.47961645970000000000 \end{bmatrix}, lambda = ,$$

-.10000000000

$$k = , 27, z = , \begin{bmatrix} 2.797762642 \\ 6.794566520 \\ -4.796164578 \\ 4.796164576 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762653191050568 \\ 0.6794566547178266080 \\ -.4796164597184658312 \\ 0.4796164595184658304 \end{bmatrix}, lambda = ,$$

-.10000000004

$$k = , 28, z = , \begin{bmatrix} -2.797762631 \\ -6.794566536 \\ 4.796164583 \\ -4.796164580 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762636595525262 \\ -.6794566549589133072 \\ 0.4796164592592329166 \\ -.4796164589592329160 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 29, z = , \begin{bmatrix} 2.797762656 \\ 6.794566495 \\ -4.796164576 \\ 4.796164572 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762669988813280 \\ 0.6794566528972832475 \\ -.4796164599980822880 \\ 0.4796164595980822860 \end{bmatrix}, lambda = ,$$

-.10000000005

$$k = , 30, z = , \begin{bmatrix} -2.797762655 \\ -6.794566520 \\ 4.796164586 \\ -4.796164585 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762660595525310 \\ -.6794566533589133040 \\ 0.4796164595592329172 \\ -.4796164594592329170 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 31, z = , \begin{bmatrix} 2.797762637 \\ 6.794566526 \\ -4.796164581 \\ 4.796164580 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762645393287911 \\ 0.6794566546383699578 \\ -.4796164595388493743 \\ 0.4796164594388493740 \end{bmatrix}, lambda = ,$$

-.10000000003

$$k = , 32, z = , \begin{bmatrix} -2.797762668 \\ -6.794566490 \\ 4.796164578 \\ -4.796164574 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762681988813340 \\ -.6794566523972832450 \\ 0.4796164601980822890 \\ -.4796164597980822870 \end{bmatrix}, lambda = ,$$

-.10000000005

$$k = , 33, z = , \begin{bmatrix} 2.797762717 \\ 6.794566459 \\ -4.796164588 \\ 4.796164586 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762730988813585 \\ 0.6794566492972832295 \\ -.4796164611980822940 \\ 0.4796164609980822930 \end{bmatrix}, lambda = ,$$

-.10000000005

$$k = , 34, z = , \begin{bmatrix} -2.797762679 \\ -6.794566520 \\ 4.796164595 \\ -4.796164598 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762681797762679 \\ -.6794566526794566520 \\ 0.4796164599796164595 \\ -.4796164602796164598 \end{bmatrix}, lambda = ,$$

-.10000000000

$$k = , 35, z = , \begin{bmatrix} 2.797762668 \\ 6.794566510 \\ -4.796164590 \\ 4.796164589 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762673595525336 \\ 0.6794566523589133020 \\ -.4796164599592329180 \\ 0.4796164598592329178 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 36, z = , \begin{bmatrix} -2.797762712 \\ -6.794566464 \\ 4.796164588 \\ -4.796164587 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762723191050848 \\ -.6794566491178265856 \\ 0.4796164607184658352 \\ -.4796164606184658348 \end{bmatrix}, lambda = ,$$

-.10000000005

$$k = , 37, z = , \begin{bmatrix} 2.797762711 \\ 6.794566474 \\ -4.796164593 \\ 4.796164595 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762719393288133 \\ 0.6794566494383699422 \\ -.4796164607388493779 \\ 0.4796164609388493785 \end{bmatrix}, lambda = ,$$

-.10000000003

$$k = , 38, z = , \begin{bmatrix} -2.797762667 \\ -6.794566531 \\ 4.796164595 \\ -4.796164597 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.27977626670000000000 \\ -.67945665310000000000 \\ 0.47961645950000000000 \\ -.47961645970000000000 \end{bmatrix}, lambda = ,$$

-.10000000000

$$k = , 39, z = , \begin{bmatrix} 2.797762642 \\ 6.794566520 \\ -4.796164578 \\ 4.796164576 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762653191050568 \\ 0.6794566547178266080 \\ -.4796164597184658312 \\ 0.4796164595184658304 \end{bmatrix}, lambda = ,$$

-.10000000004

$$k = , 40, z = , \begin{bmatrix} -2.797762631 \\ -6.794566536 \\ 4.796164583 \\ -4.796164580 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762636595525262 \\ -.6794566549589133072 \\ 0.4796164592592329166 \\ -.4796164589592329160 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 41, z = , \begin{bmatrix} 2.797762656 \\ 6.794566495 \\ -4.796164576 \\ 4.796164572 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762669988813280 \\ 0.6794566528972832475 \\ -.4796164599980822880 \\ 0.4796164595980822860 \end{bmatrix}, lambda = ,$$

-.10000000005

$$k = , 42, z = , \begin{bmatrix} -2.797762655 \\ -6.794566520 \\ 4.796164586 \\ -4.796164585 \end{bmatrix}, z\ norme = , \begin{bmatrix} -.2797762660595525310 \\ -.6794566533589133040 \\ 0.4796164595592329172 \\ -.4796164594592329170 \end{bmatrix}, lambda = ,$$

-.10000000002

$$k = , 43, z = , \begin{bmatrix} 2.797762637 \\ 6.794566526 \\ -4.796164581 \\ 4.796164580 \end{bmatrix}, z\ norme = , \begin{bmatrix} 0.2797762645393287911 \\ 0.6794566546383699578 \\ -.4796164595388493743 \\ 0.4796164594388493740 \end{bmatrix}, lambda = ,$$

-.10000000003

$$k = , 44, z = , \begin{bmatrix} -2.797762668 \\ -6.794566490 \\ 4.796164578 \\ -4.796164574 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797762681988813340 \\ -.6794566523972832450 \\ 0.4796164601980822890 \\ -.4796164597980822870 \end{bmatrix}, \lambda = ,$$

-.10000000005

$$k = , 45, z = , \begin{bmatrix} 2.797762717 \\ 6.794566459 \\ -4.796164588 \\ 4.796164586 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2797762730988813585 \\ 0.6794566492972832295 \\ -.4796164611980822940 \\ 0.4796164609980822930 \end{bmatrix}, \lambda = ,$$

-.10000000005

$$k = , 46, z = , \begin{bmatrix} -2.797762679 \\ -6.794566520 \\ 4.796164595 \\ -4.796164598 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797762681797762679 \\ -.6794566526794566520 \\ 0.4796164599796164595 \\ -.4796164602796164598 \end{bmatrix}, \lambda = ,$$

-.10000000000

$$k = , 47, z = , \begin{bmatrix} 2.797762668 \\ 6.794566510 \\ -4.796164590 \\ 4.796164589 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2797762673595525336 \\ 0.6794566523589133020 \\ -.4796164599592329180 \\ 0.4796164598592329178 \end{bmatrix}, \lambda = ,$$

-.10000000002

$$k = , 48, z = , \begin{bmatrix} -2.797762712 \\ -6.794566464 \\ 4.796164588 \\ -4.796164587 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} -.2797762723191050848 \\ -.6794566491178265856 \\ 0.4796164607184658352 \\ -.4796164606184658348 \end{bmatrix}, \lambda = ,$$

-.10000000005

$$k = , 49, z = , \begin{bmatrix} 2.797762711 \\ 6.794566474 \\ -4.796164593 \\ 4.796164595 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.2797762719393288133 \\ 0.6794566494383699422 \\ -.4796164607388493779 \\ 0.4796164609388493785 \end{bmatrix}, \lambda = ,$$

-.10000000003

-.10000000000

> # La distance minimale entre 2,5 et une valeur propre est donc de

-0,1. Ainsi, on trouve comme valeur propre $2,5 - 0,1 = 2,4$.

> # La dernière valeur propre s'obtient en rappelant que la somme des valeurs propres d'une matrice est égale à la trace de celle-ci (la trace est la somme des éléments diagonaux).

> linalg[trace](a) - (2.4 - 1 + 4.8);

-2.4

(4)

> a:=evalm((1/9)*matrix(3,3,[5,2,2,2,-1,8,2,8,-1]));z:=matrix(3,1,[1,2,0]);puiss(a,z,10,10);

$$a := \begin{bmatrix} \frac{5}{9} & \frac{2}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{-1}{9} & \frac{8}{9} \\ \frac{2}{9} & \frac{8}{9} & \frac{-1}{9} \end{bmatrix}$$

(5)

$$z := \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$k = , 0, z = , \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.4472135954 \\ 0.8944271908 \\ 0. \end{bmatrix}, \lambda = , 0$$

$$k = , 1, z = , \begin{bmatrix} 0.4472135954 \\ 0. \\ 0.8944271908 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.44721359540000000000 \\ 0. \\ 0.89442719080000000000 \end{bmatrix}, \lambda = , 0.1999999999$$

$$k = , 2, z = , \begin{bmatrix} 0.4472135954 \\ 0.8944271908 \\ 0. \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \\ 0. \end{bmatrix}, \lambda = , 0.1999999999$$

$$k = , 3, z = , \begin{bmatrix} 0.4472135954 \\ 0. \\ 0.8944271908 \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.44721359540000000000 \\ 0. \\ 0.89442719080000000000 \end{bmatrix}, \lambda = , 0.1999999999$$

$$k = , 4, z = , \begin{bmatrix} 0.4472135954 \\ 0.8944271908 \\ 0. \end{bmatrix}, z \text{ norme} = , \begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \\ 0. \end{bmatrix}, \lambda = , 0.1999999999$$


```

k = , 5, z = ,  $\begin{bmatrix} 0.4472135954 \\ 0.8944271908 \end{bmatrix}$  , z norme = ,  $\begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \end{bmatrix}$  , lambda = , 0.1999999999

k = , 6, z = ,  $\begin{bmatrix} 0.4472135954 \\ 0.8944271908 \\ 0. \end{bmatrix}$  , z norme = ,  $\begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \\ 0. \end{bmatrix}$  , lambda = , 0.1999999999

k = , 7, z = ,  $\begin{bmatrix} 0.4472135954 \\ 0.8944271908 \end{bmatrix}$  , z norme = ,  $\begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \end{bmatrix}$  , lambda = , 0.1999999999

k = , 8, z = ,  $\begin{bmatrix} 0.4472135954 \\ 0.8944271908 \\ 0. \end{bmatrix}$  , z norme = ,  $\begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \\ 0. \end{bmatrix}$  , lambda = , 0.1999999999

k = , 9, z = ,  $\begin{bmatrix} 0.4472135954 \\ 0.8944271908 \end{bmatrix}$  , z norme = ,  $\begin{bmatrix} 0.44721359540000000000 \\ 0.89442719080000000000 \end{bmatrix}$  , lambda = , 0.1999999999
0.1999999999

```

>

Comme il y a oscillation, on a $\lambda_1^2 = z_{2k}^t * A * A * z_{2k}$

```

> evalm(linalg[transpose](matrix([.44721359540000000000],
    [.89442719080000000000], [0.])))&a&a&matrix([
    [.44721359540000000000], [.89442719080000000000], [0.]));
    [0.99999999955297450580]

```

(6)

```

> sqrt(.99999999955297450580);
    0.99999999977648725288

```

(7)

On a donc $\lambda_1 = 0.99999999977648725288$ et $\lambda_2 = -0.99999999977648725288$.

Trouvons maintenant les vecteurs propres associés à ces valeurs propres.
Ce sont $z_{2k+1} \pm \lambda_1 * z_{2k_normé}$

```

> evalm(matrix([.4472135954], [0.], [.8944271908])) +
    .99999999977648725288 * matrix([.44721359540000000000],

```

```
[.8944271908000000000], [0.]));
```

$$\begin{bmatrix} 0.89442719070004206074 \\ 0.89442719060008412149 \\ 0.8944271908 \end{bmatrix}$$

(8)

```
> evalm(matrix([.4472135954], [0.], [.8944271908])) -  
.99999999977648725288 * matrix([.4472135954000000000],  
[.8944271908000000000], [0.]));
```

$$\begin{bmatrix} 9.995793926 \cdot 10^{-11} \\ -.89442719060008412149 \\ 0.8944271908 \end{bmatrix}$$

(9)

```
>
```